

Name: _____ Matriculation Number: _____

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Exam for the Lecture
Cyber-Physical Systems – Discrete Models
WS 2019/20

The exam consists of 8 sheets, printed on both sides. There are 12 exercises. If you write your solution on a separate page, note this on the page of the exercise. You can get additional sheets of paper upon request.

Write down your name and matriculation number on every sheet.

The only resources allowed are one handwritten DIN A4 sheet of paper, and a pen.

Use a permanent pen, i.e., no pencil, and no red color.

You have 90 minutes to solve the exercises.

In case of illness, you must supply proof of your illness by submitting a medical report to the Examination Office. Please note that the medical examination must be done at the latest on the same day of the missed exam. In case of illness while writing the exam please contact the supervisory staff, inform them about your illness and immediately see your doctor. The medical certificate must be submitted within 3 days.

A student who disrupts the orderly proceedings of an examination will be excluded from the remainder of the exam by the respective examiners or invigilators. In such a case, the written exam of the student in question will be graded as “nicht bestanden” (5.0, fail) on the grounds of cheating. In severe cases, the Board of Examiners will exclude the student from further examinations.

Exercise	Points
1. General Questions about Modeling Parallel Systems	of 6
2. Hardware Circuit	of 12
3. Simple Traffic Light System	of 14
Bonus exercise	of 2*
4. Locking Protocol	of 18
5. Paths and Traces	of 6
6. Linear Time Properties	of 9
7. Safety and Liveness Properties	of 10
8. Decomposition of Linear Time Properties	of 5
9. –	of 6
10. –	of 6
11. Regular Expressions, Omega-Regular Expressions, Büchi Automata	of 8
12*. Bonus: General Questions about Safety and Liveness Properties	of 5*
Total Points	of 100

Exercise 1 (General Questions about Modeling Parallel Systems)

6

A parallel system is composed of processes which communicate with each other in order to synchronize themselves. There are several forms of parallel composition which differ in the communication mechanism, i.e., in the way the processes communicate with each other in order to synchronize each other. For each task of synchronization, there is an appropriate communication mechanism.

Consider a parallel system composed of Process 1 and Process 2. What is an appropriate communication mechanism ...

- (a) if we want that Process 1 moves from s_1 to s'_1 simultaneously (in the same transition) when Process 2 moves from s_2 to s'_2 ?
- (b) if we want to forbid that Process 2 can make a transition whenever Process 1 is in a given state?
- (c) if the two processes work together in a producer-consumer fashion, e.g., one process produces pieces of information that the other process consumes, or one process produces tasks for the other process?

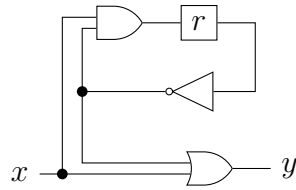
For each case, give an informal explanation. You can also use a drawing to illustrate your explanation.

Exercise 2

(Hardware Circuit)

12

Consider the following sequential hardware circuit.



Provide the labeled transition system of this hardware circuit (i.e., states are labeled by sets of atomic propositions, transitions are not labeled). The states are the evaluations of the input x and the register r . The transitions represent the stepwise behavior of the circuit. The values of the input x change nondeterministically. The atomic propositions $\{X, Y, R\}$ stand for $x = 1$, $y = 1$ and $r = 1$, respectively. Initially the register r has the value 0 (**false**).

For your reference: $\square \triangleright$ = AND gate, $\sqcup \triangleright$ = OR gate, $\triangleright \circ$ = NOT gate

Exercise 3

(Simplified Traffic Light System)

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This exercise is about the design of a traffic light system. The system consists of two simple traffic lights that can switch only between the colors red and green, and one controller. The three components are synchronized via handshaking. The system consisting of just one traffic light switches nondeterministically between the colors red and green; initially, the light is red. The system consisting of just the two traffic lights (without the controller) switches nondeterministically between combinations of red and green. All combinations of red and green are reachable in this system. There is no synchronization between the two traffic lights. In the system consisting of the two traffic lights and the controller, each of the traffic lights can switch between red and green, but the traffic lights are never both green.

- (a) Draw the transition systems TS_1 and TS_2 for each of the two traffic lights. Do not forget to label the edges; the edge labels will be used for synchronization in (c) and (d).
- (b) Draw the transition system that arises from the parallel composition of the transition systems TS_1 and TS_2 . Do not forget to label the edges.
- (c) Draw the transition system TS_0 for the controller. Do not forget to label the edges.
- (d) Draw the transition system $TS_0 || (TS_1 || TS_2)$ that arises from the parallel composition of the two transition systems TS_0 and $TS_1 || TS_2$. Do not forget to label the edges.
- (e) (Bonus exercise)
Explain why it is not possible to have a transition from the combination (green, red) to the combination (red, green) in the system $TS_1 || TS_2$.

2*

Give only the reachable part of a transition system. Do not forget to mark the initial state.

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Exercise 4

(Locking Protocol)

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Consider the following locking protocol. The initial value of the variable *lock* is 0.

Process 1:	Process 2:
while true do <i>nc:</i> ...noncritical section ... await (<i>lock</i> $\neq$ 1) do <i>lock</i> := 1; od <i>c:</i> ...critical section ... <i>lock</i> := 2; od	while true do <i>nc:</i> ...noncritical section ... await (<i>lock</i> $\neq$ 1) do <i>lock</i> := 2; od <i>c:</i> ...critical section ... <i>lock</i> := 0; od

- (a) Draw the program graphs PG_1 and PG_2 for Program 1 and Program 2. The nodes of the program graphs are nc_1 and c_1 , and nc_2 and c_2 , respectively. The await statement is executed atomically, i.e., the check for $lock \neq 1$ and the update of *lock* are executed in one step.
- (b) Draw the program graph for the parallel composition of Program 1 and Program 2.
- (c) Draw the reachable part of the transition system TS for the parallel composition of the programs.
- (d) Does the protocol satisfy the mutual-exclusion property? Explain your answer in sufficient detail.

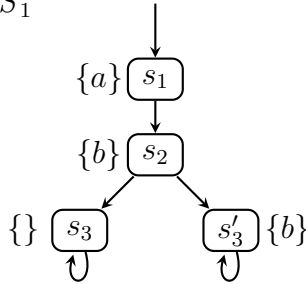
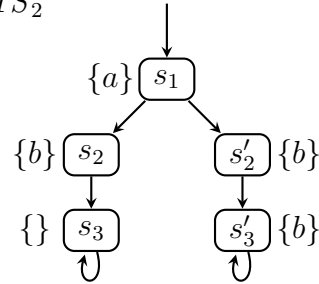
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Exercise 5

(Paths and Traces)

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Consider the following two transition systems.

 TS_1  TS_2 

In this exercise, we consider *infinite* paths and *infinite* traces.
Solve the following tasks.

- Give the set of all paths that are paths of TS_1 but not of TS_2 .
- Give the set of all paths that are paths of TS_2 but not of TS_1 .
- Give the set of all traces that are traces of TS_1 but not of TS_2 .
- Give the set of all traces that are traces of TS_2 but not of TS_1 .
- Does there exist an LT-property that is satisfied by TS_1 but not by TS_2 ? If so, give an example of such an LT-property. If not, explain why this is the case.

If one of the sets is empty, please indicate this by writing “ $\{\}$ ”.

Exercise 6

(Linear Time Properties)

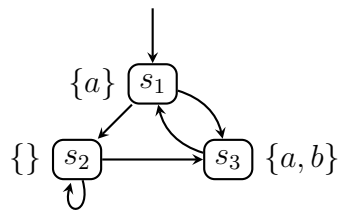
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Let $AP = \{a, b\}$. Consider the following properties.

- (P_1) Always (at any point of time) a holds.
- (P_2) At least once, a holds.
- (P_3) Every time a holds, there will eventually be a point of time where b holds.

For each of the properties above,

- (a) formalize the property as a set of traces (using set comprehension in the style $P = \{A_0A_1A_2 \dots \mid \forall i. b \in A_i \wedge \exists j \leq i. b \notin A_j\}$),
- (b) state whether or not the transition system below satisfies the property (explain your answer!).



Exercise 7

(Safety and Liveness Properties)

10

Consider the set $AP = \{a, b, c, d\}$ of atomic propositions.

Determine which of the following properties are safety properties, which are liveness properties, which are none, and which are both. Explain your answer using the definition of safety or liveness properties.

- (a) A state with the label $\{a, b\}$ must never be reached.
- (b) Every occurrence of a must directly be preceded by c .
- (c) Eventually a holds, and eventually a does not hold.
- (d) Every occurrence of d must eventually be followed by b .
- (e) Every occurrence of c is directly followed by a or it is not.

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Exercise 8

(Decomposition of Linear Time Properties)

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For the following linear time properties P , give a safety property P_{safe} and a liveness property P_{live} such that $P = P_{safe} \cap P_{live}$.

- (a) P is the property: at exactly 0 points of time, a holds.
- (b) P is the property: at exactly 2 points of time, a holds.

Use informal descriptions of P_{safe} and P_{live} in the style above.

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Exercise 9

(Fairness and Traces)

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The content of this exercise was not covered this semester.

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Exercise 10

(Fairness Assumptions)

6

The content of this exercise was not covered this semester.

Exercise 11 (Regular Expressions, Omega-Regular Expressions, Büchi Automata)

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Let $AP = \{a, b\}$ be a set of atomic propositions. Consider the following property P :

*“every time a holds, there will eventually be a later point in time
where a does **not** hold and b does **not** hold”*

Complete the tasks below.

- (a) The content of this sub-exercise was not covered this semester.
- (b) Draw a Büchi automaton that accepts exactly the traces that satisfy the property P . Use symbolic notation.
- (c) The content of this sub-exercise was not covered this semester.
- (d) The content of this sub-exercise was not covered this semester.

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Exercise 12 (Bonus: General Questions about Safety and Liveness Properties)

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For each of the following tasks, give linear time properties P_1 and P_2 such that $P_1 \subseteq P_2$. If no such properties exist, give an explanation why this is the case.

- (a) P_1 is a safety property, and P_2 is a liveness property.
- (b) P_1 is a liveness property, and P_2 is a safety property.
- (c) P_1 is a safety property, and P_2 is *not* a safety property.
- (d) P_1 is a liveness property, and P_2 is *not* a liveness property.
- (e) P_1 is *not* an invariant, but P_2 is an invariant.

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